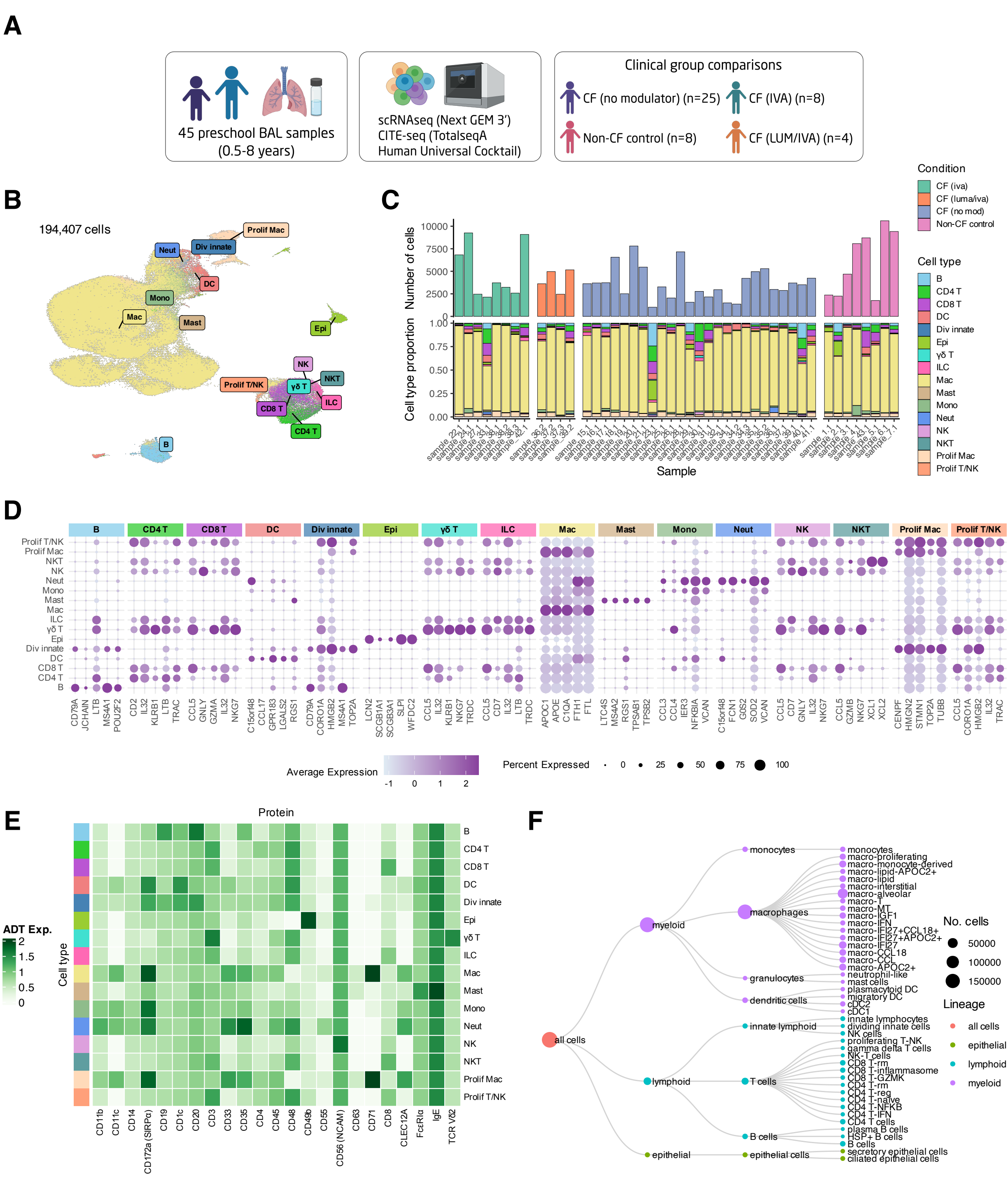
A
45 preschool BAL samples
(0.5-8 years)
scRNAseq (Next GEM 3')
CITE-seq (TotalseqA
Human Universal Cocktail)
Clinical group comparisons
CF (no modulator) (n=25)
CF (IVA) (n=8)
Non-CF control (n=8)
CF (LUM/IVA) (n=4)
B
194,407 cells
Prolif Mac
Neut
Div innate
DC
Mono
Mac
Mast
Epi
NK
NKT
Prolif T/NK
γδ T
ILC
CD8 T
CD4 T
B
C
Cell type proportion
Number of cells
10000
7500
5000
2500
0
1.00
0.75
0.50
0.25
0.00
Sample
Condition
CF (iva)
CF (luma/iva)
CF (no mod)
Non-CF control
Cell type
B
CD4 T
CD8 T
DC
Div innate
Epi
γδ T
ILC
Mac
Mast
Mono
Neut
NK
NKT
Prolif Mac
Prolif T/NK
D
B
CD4 T
CD8 T
DC
Div innate
Epi
γδ T
ILC
Mac
Mast
Mono
Neut
NK
NKT
Prolif Mac
Prolif T/NK
CD79A JCHAIN LTB MS4A1 POU2F2
CD2 IL32 KLRB1 LTB TRAC
CCL5 GNLY GZMA IL32 NKG7
C15orf48 CCL17 GPR183 LGALS2 RGS1
CD79A CORO1A HMGB2 MS4A1 TOP2A
LCN2 SCGB1A1 SCGB3A1 SLPI WFDC2
CCL5 IL32 KLRB1 NKG7 TRDC
CCL5 CD7 IL32 LTB TRDC
APOC1 APOE C1QA FTH1 FTL
LTC4S MS4A2 RGS1 TPSAB1 TPSB2
CCL3 CCL4 IER3 NFKBIA VCAN
C15orf48 FCN1 G0S2 SOD2 VCAN
CCL5 CD7 GNLY IL32 NKG7
CCL5 GZMB NKG7 XCL1 XCL2
CENPF HMGN2 STMN1 TOP2A TUBB
CCL5 CORO1A HMGB2 IL32 TRAC
Average Expression
-1 0 1 2
Percent Expressed
0 25 50 75 100
E
Protein
ADT Exp.
2
1.5
1
0.5
0
Cell type
B
CD4 T
CD8 T
DC
Div innate
Epi
γδ T
ILC
Mac
Mast
Mono
Neut
NK
NKT
Prolif Mac
Prolif T/NK
CD11b CD11c CD14 CD172a (SIRPα) CD19 CD1c CD20 CD3 CD33 CD35 CD4 CD45 CD48 CD49b CD55 CD56 (NCAM) CD63 CD71 CD8 CLEC12A FcεRIα IgE TCR Vδ2
F
all cells
myeloid
lymphoid
epithelial
monocytes
macrophages
granulocytes
dendritic cells
innate lymphoid
T cells
B cells
epithelial cells
monocytes
macro-proliferating
macro-monocyte-derived
macro-lipid-APOC2+
macro-lipid
macro-interstitial
macro-alveolar
macro-T
macro-MT
macro-IGF1
macro-IFN
macro-IFI27+CCL18+
macro-IFI27+APOC2+
macro-IFI27
macro-CCL18
macro-CCL
macro-APOC2+
neutrophil-like
mast cells
plasmacytoid DC
migratory DC
cDC2
cDC1
innate lymphocytes
dividing innate cells
NK cells
proliferating T-NK
gamma delta T cells
NK-T cells
CD8 T-rm
CD8 T-inflammasome
CD8 T-GZMK
CD4 T-rm
CD4 T-reg
CD4 T-naive
CD4 T-NFKB
CD4 T-IFN
CD4 T cells
plasma B cells
HSP+ B cells
B cells
secretory epithelial cells
ciliated epithelial cells
No. cells
50000
100000
150000
Lineage
all cells
epithelial
lymphoid
myeloid